

1st REVIEW

WarehouseIQ: Smart Warehouse & Inventory Management System

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Under the Guidance
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GROUP NUMBER:4

Name of Candidate	Role
Yashas	MySQL, ER Diagram, Tables, SQL, Inventory, Stock Logic, Movement History, Audit & Reports
Gnaneshwar	React, CSS, Login, Dashboard, Products, Suppliers, Warehouse, Inventory, Stock IN/OUT & Reports
Venkatesh	Flask, MySQL Connection, REST APIs, Login, Product/Supplier/Warehouse/Inventory/Stock/Movement/Report APIs & Postman

ABSTRACT

- **WarehouseIQ** is a Smart Warehouse and Inventory Management System designed to efficiently manage warehouse operations and inventory data through a centralized database system.
- The system manages **products, suppliers, warehouses, inventory levels, stock-in, stock-out, and stock movement history**. It helps maintain accurate stock information, identify low-stock products, track inventory movements, and generate useful reports.
- WarehouseIQ reduces dependency on manual records and scattered spreadsheets by providing a structured and reliable system for warehouse data management. The project uses **React for the frontend, Flask for the backend, and MySQL as the database**.

INTRDOUCTION

Warehouse and inventory management involves maintaining information about products, suppliers, storage locations, stock quantities, and inventory movements.

Traditional methods often depend on **manual records and spreadsheets**, making it difficult to maintain accurate and updated inventory information.

WarehouseIQ provides a centralized platform where warehouse-related data can be stored, managed, and accessed efficiently.

- **The system provides:**

- Product management

- Supplier management

- Warehouse management

- Inventory tracking

- Stock In / Stock Out

- Movement history

- Dashboard and reports

EXISTING SYSTEM

- **Existing System**

1. Many warehouses currently rely on:
2. Manual record keeping
3. Excel and spreadsheet-based inventory
4. Separate files for different warehouse operations
5. Manual stock updates
6. Manual report preparation

- **Limitations**

1. Stock information may become inaccurate
2. Difficult to track stock movements
3. Low-stock products may be missed
4. Data is scattered across multiple files
5. Reports take more time to prepare
6. Higher possibility of human errors
7. Difficult to maintain complete movement history
8. These limitations can lead to slower decisions and operational risks.

PROBLEM

The main problem is the **lack of a centralized and reliable warehouse inventory management system.**

When product, supplier, warehouse, and stock information is maintained separately, it becomes difficult to obtain an accurate view of current inventory.

- **Major Problems**

1. Inaccurate stock records
2. Difficulty monitoring inventory levels
3. Missing or delayed low-stock alerts
4. Difficult stock-in and stock-out tracking
5. No centralized movement history
6. Slow report generation
7. Data duplication and inconsistency

SOLUTION(Contribution)

- WarehouseIQ provides a **centralized database-driven solution** for warehouse and inventory management.
- The system:
 1. Stores product and SKU information centrally
 2. Maintains supplier records
 3. Manages warehouse and storage information
 4. Tracks available and reserved inventory
 5. Records stock entering and leaving the warehouse
 6. Maintains stock movement history
 7. Identifies low-stock products
 8. Provides dashboards and reports

- **Contribution**

The project improves **inventory visibility, data consistency, traceability, and operational efficiency** by bringing major warehouse operations into a single system.

Objectives

- **Primary Objectives**

1. To develop a centralized warehouse inventory management system.
2. To maintain accurate product and inventory records.
3. To manage supplier and warehouse information efficiently.
4. To track stock-in and stock-out operations.
5. To maintain complete stock movement history.
6. To identify products that require stock replenishment.
7. To reduce manual errors in inventory management.
8. To provide dashboards and reports for better decision-making.
9. To design a normalized and reliable MySQL database.
10. To provide secure, role-based access to the system.

Modules

- **Core Modules**
- **01. Login & Authentication**
Secure user login and role-based access.
- **02. Product Management**
Manage product details, SKUs, categories, and product records.
- **03. Supplier Management**
Store supplier profiles, contact information, and lead-time details.
- **04. Warehouse Management**
Manage warehouse locations, storage bins, and capacity information.
- **05. Inventory Management**
Track stock quantities, available stock, reserved stock, and inventory valuation.
- **06. Stock In / Stock Out**
Record receiving and dispatch operations with proper validation.
- **07. Movement History**
Maintain records of all inventory movements for tracking and auditing.
- **08. Dashboard & Reports**
Display important inventory information and generate operational reports.

SOFTWARE REQUIREMENTS

- **Frontend**

1. **React.js** – Used to build the interactive and responsive user interface.
2. **HTML** – Provides the basic structure of the application pages.
3. **CSS** – Used for styling, layout, and responsive design.
4. **JavaScript** – Handles client-side functionality and user interactions.

- **Backend**

1. **Python** – Used for implementing the backend application logic.
2. **Flask** – Lightweight framework used to develop the backend and RESTful services.
3. **RESTful API** – Enables communication between the React frontend and backend.

- **Database**

1. **MySQL** – Stores products, suppliers, customers, inventory, purchases, sales, and warehouse transaction data.
2. **Normalized Relational Database** – Organizes data into related tables to reduce redundancy and maintain data consistency.
3. **SQL** – Used for inserting, updating, deleting, and retrieving warehouse and inventory information.

- **Development Tools**

1. **Visual Studio Code** – Used for coding and project development.
2. **Git/GitHub** – Used for version control and collaborative project management.
3. **Postman** – Used for testing and validating RESTful APIs.

- **Operating System**

1. **Windows / Linux / macOS** – Supported development and execution environments.

CONCLUSION AND FUTURE SCOPE

- **Conclusion**

WarehouseIQ provides a centralized and structured solution for managing warehouse and inventory operations.

The system helps organizations maintain accurate inventory records, track stock movements, manage products and suppliers, monitor warehouse information, and generate useful reports.

By replacing scattered manual records with a centralized database system, WarehouseIQ can improve **inventory visibility, accuracy, traceability, and operational efficiency.**

- **Future Scope**

The system can be further enhanced with:

1. Automated low-stock notifications
2. Barcode/QR-code support
3. Mobile-friendly interface
4. Improved inventory reports
5. Stock usage analytics
6. Automated restock suggestions

THANK YOU